LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Measurement of various electrical and electronic parameters using an Oscilloscope.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Measurement of various electrical and electronic parameters using an Oscilloscope.
5. Specific Objectives	24.8.1.1 Trainee will be able to measure various parameters using an Oscilloscope.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Hands-on measurement of voltage, frequency, and duty cycle using a digital oscilloscope; Safety: High voltage caution, proper grounding, and probe compensation verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an oscilloscope?
(b) How does a digital oscilloscope differ from an analog one?
(c) What safety precautions are required when connecting probes to live circuits?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Oscilloscope basic controls and probe selection • Vertical sensitivity and time base settings • Triggering modes and waveform stability	L & Ds	PPT, IB	15 min
(b)	• Measuring peak-to-peak voltage and RMS values • Determining frequency and period from waveforms • Calculating duty cycle for pulse signals	L & Ds	PPT, PDF	20 min
(c)	• Probe compensation and ground lead effects • Error sources in high-frequency measurements • Documentation of measurement results	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between vertical sensitivity and measured voltage.
(b) Describe how to adjust the trigger level to stabilize a noisy waveform.
(c) Why is probe compensation necessary before taking accurate measurements?

12. Home Assignment and Follow Up	Prepare a brief report on the procedure to measure the breakdown voltage of a diode using an oscilloscope, referencing the next lesson plan (LP 10).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Measurement of Breakdown voltage & potential barrier of a diode.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min